

MongoDB Complete Notes

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1. What is MongoDB?

MongoDB is a NoSQL database that stores data in the form of documents. Documents are stored in collections and collections are stored inside databases.

2. Show Databases

Command to display all databases:

```
show dbs
```

3. Create / Switch Database

Use database command:

```
use Abeeha
```

```
Check current database:  
db
```

4. Create Collection

Create a collection using:

```
db.createCollection("students")
```

```
Check collections:  
show collections
```

5. Insert One Document

Insert a single document:

```
db.students.insertOne({
```

```
name: "Ali",  
age: 21,  
city: "Sukkur"  
})
```

6. Insert Many Documents

Insert multiple records:

```
db.students.insertMany([  
  {name: "Ali", age: 21, city: "Sukkur"},  
  {name: "Ahmed", age: 22, city: "Karachi"},  
  {name: "Abeeha", age: 20, city: "Larkana"},  
  {name: "Hamdan", age: 25, city: "Hyderabad"}  
)
```

7. Find Operations

Find all documents:

```
db.students.find()  
  
db.students.find().pretty()
```

8. Find One Document

Get first matching document:

```
db.students.findOne()
```

9. Comparison Operators

MongoDB comparison operators are used for filtering

data. **10. Greater Than (\$gt)**

Age greater than 20:

```
db.students.find({age: {$gt: 20}})
```

11. Greater Than Equal (\$gte)

Age greater than or equal to 20:

```
db.students.find({age:{$gte:20}})
```

12. Less Than (\$lt)

Age less than 20:

```
db.students.find({age:{$lt:20}})
```

13. Less Than Equal (\$lte)

Age less than or equal to 20:

```
db.students.find({age:{$lte:20}})
```

14. Between Values

Age between 20 and 25:

```
db.students.find({age:{$gte:20,$lte:25}})
```

15. Logical Operators

AND and OR operators combine multiple conditions.

AND Example

```
db.students.find({age:{$gt:20},city:"Karachi"})
```

OR Example

```
db.students.find({$or:[{city:"Karachi"},{city:"Sukkur"}]}))
```

16. Not Equal (\$ne)

```
db.students.find({city:{$ne:"Karachi"}})
```

17. IN Operator (\$in)

```
db.students.find({city:{$in:["Karachi","Sukkur"]}})
```

18. NOT IN Operator (\$nin)

```
db.students.find({city:{$nin:["Karachi","Sukkur"]}})
```

19. Sorting

Ascending uses 1 and descending uses -1:

```
db.students.find().sort({age:1})
```

```
db.students.find().sort({age:-1})
```

20. Limit and Skip

Limit controls records and skip ignores records:

```
db.students.find().limit(2)
```

```
db.students.find().skip(2)
```

21. Count Documents

```
db.students.countDocuments()
```

22. Select Specific Fields

```
db.students.find({}, {name:1, age:1, _id:0})
```

23. Update One Document

```
db.students.updateOne({name:"Ali"}, {$set:{age:24}})
```

24. Update Many Documents

```
db.students.updateMany({}, {$set:{status:"Active"}})
```

25. Delete Operations

Delete one document:

```
db.students.deleteOne({name:"Ali"})
```

Delete many:

```
db.students.deleteMany({city:"Karachi"})
```

Delete all:

```
db.students.deleteMany({})
```

26. Drop Collection and

Database `db.students.drop()`

```
db.dropDatabase()
```

Assignment

1. Create school database.
2. Create students collection.
3. Insert 10 records using insertMany().
4. Find age greater than 20.
5. Find age less than 25.
6. Find age between 20 and 30.
7. Find Karachi or Sukkur students.
8. Apply sorting.
9. Update a record.
10. Delete a record.